

RealEye Software Requirements Specification

CMSI 4071 Senior Group Project

Team: Charlize, Tara, Anastasia, Anastasia, Noa Sasover

September 30, 2026

This specification defines the required behavior, measurable acceptance targets, and development and execution environments for RealEye. It covers the Chrome extension, page interaction, hosted analysis and report services, and model evaluation.

Requirements language: "shall" identifies a mandatory, testable obligation. Acceptance targets describe required outcomes; they do not assert that the system has already achieved them.

Functional requirements: 193 across four Computer Software Components.

Table of Contents

5.0 Requirements Specification	4
5.1 Introduction	4
5.1.1 Terms and Result States	4
5.2 CSCI Component Breakdown	5
5.2.1 Extension Interface CSC	5
5.2.2 Page Interaction CSC	5
5.2.3 Server CSC	5
5.2.4 Evaluation CSC	6
5.3 Functional Requirements by CSC	7
5.3.1 Extension Interface CSC	7
5.3.2 Page Interaction CSC	10
5.3.3 Server CSC	12
5.3.4 Evaluation CSC	16
5.4 Performance Requirements by CSC	18
5.4.1 Scan Button Acknowledged Within 250 ms	18
5.4.2 Up to 150 Items Enumerated Within 3 Seconds	18
5.4.3 Selected Item Result Within 10 Seconds	18
5.4.4 Full Page Progress Within 5 Seconds	18
5.4.5 Thirty Item Page Completed Within 30 Seconds	18
5.4.6 Timed Out Evidence Reported Within 5.25 Seconds	18
5.4.7 Canceled Results Removed Within 1 Second	18
5.4.8 Text Classifier F1 of at Least 0.75	18
5.4.9 Image Classifier F1 of at Least 0.75	18
5.4.10 False Positive Rates Reported by Modality	19
5.4.11 Core Workflow Completed by Keyboard	19
5.4.12 Contrast Ratios Meet WCAG 2.2 AA	19
5.4.13 Interactive Targets at Least 24 CSS Pixels	19

5.4.14 Controls Remain Accessible at 200 Percent Zoom	19
5.4.15 Shared Report Resolves Within 5 Seconds	19
5.4.16 Share Link Valid for Seven Days	19
5.4.17 Expired Snapshot Deleted Within 24 Hours	19
5.4.18 Five Concurrent Single Item Scans	19
5.4.19 Failure Feedback Within 35 Seconds	19
5.4.20 Text False Positive Rate at Most 10 Percent	19
5.4.21 Image False Positive Rate at Most 10 Percent	19
5.5 Project Environment Requirements	20
5.5.1 Development Environment Requirements	20
5.5.2 Deployment and Execution Environment	21
5.5.3 Input Limits and Reference Conditions	22
5.5.4 Deferred Extensions	23

5.0 Requirements Specification

5.1 Introduction

RealEye is a Chrome extension that helps internet users, particularly older adults and people with limited technical experience, assess whether webpage text and images may be AI generated. Users request full-page or selected-content scans, receive separate machine-learning, C2PA provenance, and accessible SynthID findings, and inspect inline warnings, explanations, and copy or share actions. The system estimates content origin and does not establish truth, safety, or fraud. Section 5.2 describes the CSCI, CSCs, CSUs, and modules; Section 5.3 specifies functional behavior by CSC; Section 5.4 defines performance requirements; and Section 5.5 identifies development and execution environments.

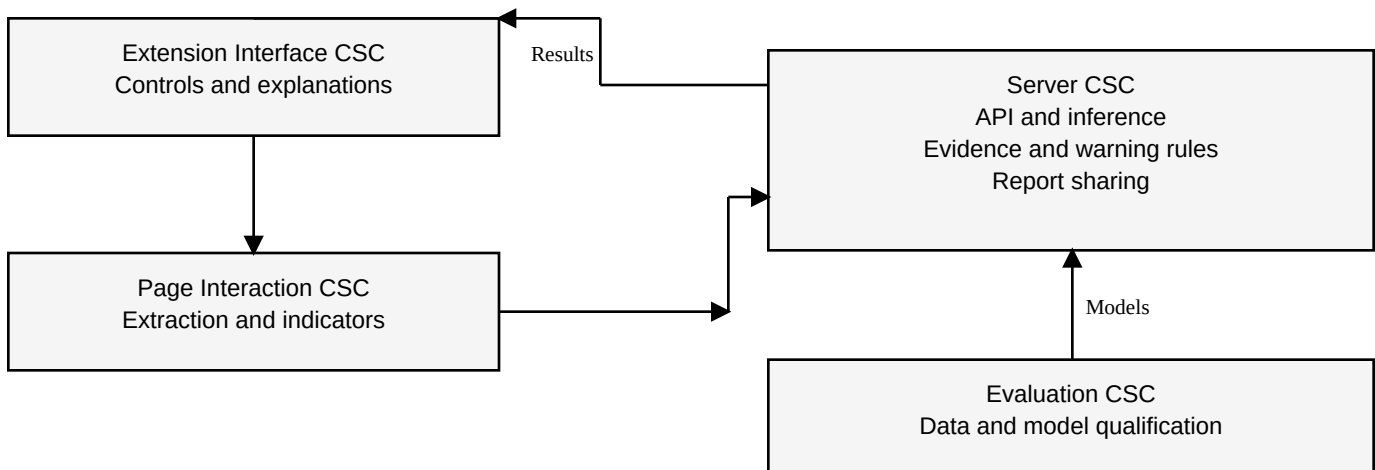


Figure 1. Scan flow and the independent model-release path.

5.1.1 Terms and Result States

Full page means currently loaded, accessible content within the scan limits; it excludes infinite scrolling and inaccessible frames. Selected content means highlighted text or a user-chosen image. Extractable text excludes OCR. A model percentage is an estimate, not a statistical confidence level. Input limits and the qualification profile are defined in Section 5.5.3.

Overall-state precedence is: Warning if any warning trigger is present; otherwise Inconclusive if at least one applicable check completes; otherwise Unable to analyze. Warning triggers are a classifier estimate at or above its modality threshold, a verified C2PA assertion of AI generation, or a positive authorized SynthID finding. Negative or missing provenance and watermark findings never cancel a warning trigger. A completed negative watermark check or completed C2PA inspection finding no credential counts as a completed check; an invalid credential, unavailable method, error, or not-applicable status alone does not.

Signal states distinguish detected, not detected, verified AI assertion, verified credential without an AI assertion, missing credential, invalid credential, not applicable, unavailable, and error where relevant. A conflict flag identifies a warning trigger accompanied by a below-threshold classifier estimate; it does not change the Warning state. A verified credential without an AI assertion is neutral on AI origin, so its combination with an above-threshold classifier estimate is not a conflict. Neither missing credentials nor absence of an AI assertion proves human authorship.

5.2 CSCI Component Breakdown

The RealEye Computer Software Configuration Item (CSCI) is divided into four Computer Software Components (CSCs), or subsystems. The Extension Interface CSC controls the extension popup, scan controls, and display of results and explanations. The Page Interaction CSC selects and extracts webpage text and images and displays inline warnings associated with analyzed content. The Server CSC processes analysis requests, runs machine-learning models, checks available C2PA credentials and SynthID watermarks, interprets evidence, and hosts shared reports. The Evaluation CSC prepares labeled datasets, fine-tunes pretrained models, and measures their performance before release. Each CSC comprises Computer Software Units (CSUs) with individually identified modules. Module names describe responsibilities rather than prescribed source files or classes.

5.2.1 Extension Interface CSC

Presents scan controls, progress, explanations, accessibility, and copy or share controls.

5.2.1.1 *Popup and Controls CSU*

Provides the scan entry point and lifecycle controls.

5.2.1.1.1 Toolbar module: Presents the RealEye icon and popup entry point.

5.2.1.1.2 Scan scope module: Selects scan scope and requests user-approved image-origin access.

5.2.1.1.3 Progress and cancellation module: Tracks active scans and handles cancellation.

5.2.1.1.4 Help module: Explains limitations, permissions, and use.

5.2.1.2 *Results and Accessibility CSU*

Presents item evidence and accessible result actions.

5.2.1.2.1 Item explanation module: Shows separate findings and overall states.

5.2.1.2.2 Status and keyboard access module: Provides announcements, focus, and keyboard operation.

5.2.1.2.3 Copy and share controls module: Previews and copies findings or requests a report link.

5.2.2 Page Interaction CSC

Extracts supported page items and associates results with inline indicators.

5.2.2.1 *Selection and Extraction CSU*

Forms eligible text and image scan items.

5.2.2.1.1 Text selection module: Obtains highlighted or discovered text.

5.2.2.1.2 Image selection module: Retrieves original image bytes through the extension service worker using granted origin permissions.

5.2.2.1.3 Content discovery and deduplication module: Applies eligibility, order, limits, and duplicate removal.

5.2.2.2 *Page Annotation CSU*

Associates warnings with page content.

5.2.2.2.1 Item location module: Locates each scanned page element.

5.2.2.2.2 Warning indicator module: Adds dismissible labels linked to explanations.

5.2.2.2.3 Stale result cleanup module: Removes obsolete indicators and results.

5.2.3 Server CSC

Hosts analysis, inference, evidence checks, warning interpretation, and shared reports.

5.2.3.1 Analysis API CSU

Validates and coordinates requests and responses.

5.2.3.1.1 Request validation module: Checks input limits, origin policy, and quotas.

5.2.3.1.2 Analysis coordination module: Dispatches independent evidence checks.

5.2.3.1.3 Response formatting module: Returns item identities, estimates, and statuses.

5.2.3.2 Text and Image Inference CSU

Runs the released classifiers.

5.2.3.2.1 Text inference module: Produces estimates for supported English text.

5.2.3.2.2 Image inference module: Produces estimates for supported decoded images.

5.2.3.2.3 Model version module: Identifies deployed artifacts and preprocessing.

5.2.3.3 C2PA and Accessible SynthID CSU

Checks available provenance and watermarks.

5.2.3.3.1 Credential verification module: Validates original-asset credentials and extracts assertions.

5.2.3.3.2 Watermark availability and detection module: Checks authorized detector availability and results.

5.2.3.3.3 Signal status module: Preserves method status, source, and limitations.

5.2.3.4 Warning Interpretation CSU

Applies the versioned warning rule.

5.2.3.4.1 Threshold module: Compares estimates with validated thresholds.

5.2.3.4.2 Evidence interpretation module: Applies state precedence and conflict flags.

5.2.3.4.3 Explanation module: Records the evidence supporting the overall state.

5.2.3.5 Sharing and Report Viewer CSU

Creates and serves limited-lifetime reports.

5.2.3.5.1 Snapshot creation module: Stores only the user-approved report fields.

5.2.3.5.2 Report access module: Renders valid links and unavailable states.

5.2.3.5.3 Expiry and deletion module: Enforces seven-day access and storage cleanup.

5.2.4 Evaluation CSC

Prepares labeled data, fine-tunes pretrained models, and qualifies release artifacts.

5.2.4.1 Dataset CSU

Manages labeled evaluation data.

5.2.4.1.1 Source and license record module: Records dataset sources and permitted uses.

5.2.4.1.2 Partition and duplicate isolation module: Separates training, validation, and test examples.

5.2.4.2 Training CSU

Produces model artifacts from pretrained models.

5.2.4.2.1 Pretrained model configuration module: Records base models and training configuration.

5.2.4.2.2 Fine tuning module: Fits classifiers on the training partition.

5.2.4.2.3 Artifact export module: Saves inference-ready models and versions.

5.2.4.3 Evaluation CSU

Qualifies thresholds and released models.

5.2.4.3.1 Held-out testing module: Measures performance on untouched test data.

5.2.4.3.2 Threshold validation module: Selects thresholds using validation data only.

5.2.4.3.3 Metrics reporting module: Records quality, class counts, and qualification conditions.

5.3 Functional Requirements by CSC

The completed system analyzes supported text and images on demand, retains independent evidence, displays warnings and explanations, and provides copy and hosted-share actions without user accounts. Each numbered statement below defines one verifiable obligation. Client display requirements and server response requirements remain separate where they have different observable outputs.

5.3.1 Extension Interface CSC

The Extension Interface CSC will allow users to choose a scan scope, start or cancel scans, and view progress. It will present results and explanations, provide help, and support accessible controls for reviewing, copying, and sharing individual item findings.

5.3.1.1 Toolbar access The Extension Interface CSC shall display a RealEye control accessible from the Chrome toolbar after installation.

5.3.1.2 Scan action The Extension Interface CSC shall present a button labeled "Start scan" for the current page.

5.3.1.3 Scan choices The Extension Interface CSC shall present separate choices for "Full page" and "Selected content" scans.

5.3.1.4 Scope explanation The Extension Interface CSC shall display the content types and page scope included in each scan choice before the user starts a scan.

5.3.1.5 Explicit initiation The Extension Interface CSC shall start a scan only after the user activates the "Start scan" button.

5.3.1.6 Active tab The Extension Interface CSC shall identify the active tab to which a requested scan applies.

5.3.1.7 Restricted page control The Extension Interface CSC shall disable the start action when Chrome does not allow access to the active page.

5.3.1.8 Restricted page explanation The Extension Interface CSC shall explain when a page cannot be scanned because extension access is unavailable.

5.3.1.9 Scan progress The Extension Interface CSC shall display a "Scanning..." status after the user starts a scan and while the scan remains active.

- 5.3.1.10 Item count** The Extension Interface CSC shall show the number of supported items found after extraction completes.
- 5.3.1.11 Cancel scan** The Extension Interface CSC shall provide a cancel action for a scan that is still running.
- 5.3.1.12 End progress** The Extension Interface CSC shall stop displaying progress when a scan completes, fails, or is canceled.
- 5.3.1.13 Completion summary** The Extension Interface CSC shall display a completion summary containing counts of analyzed, flagged, and unavailable items.
- 5.3.1.14 New scan** The Extension Interface CSC shall let the user start a new scan after an earlier scan completes or fails.
- 5.3.1.15 Stale results** The Extension Interface CSC shall label an earlier result as “Stale” when its recorded page URL or page state no longer matches the active page.
- 5.3.1.16 Help view** The Extension Interface CSC shall provide a help view defining the overall result states, classifier uncertainty, and meanings of unavailable or missing evidence signals.
- 5.3.1.17 In page results** The Extension Interface CSC shall display the current result without requiring navigation to an external detection website.
- 5.3.1.18 Dismiss panel** The Extension Interface CSC shall provide a button labeled "Close results" that dismisses the results panel.
- 5.3.1.19 Readable controls** The Extension Interface CSC shall display complete control labels without overlap or clipping at popup widths from 320 to 600 CSS pixels.
- 5.3.1.20 Item type** The Results and Accessibility CSU shall identify whether the analyzed item is text or an image.
- 5.3.1.21 Separate model estimate** The Results and Accessibility CSU shall display the classifier estimate for the selected item separately from provenance and watermark findings.
- 5.3.1.22 Estimate label** The Results and Accessibility CSU shall label every classifier percentage "Model estimate".
- 5.3.1.23 C2PA result** The Results and Accessibility CSU shall display the C2PA finding as a separate labeled result.
- 5.3.1.24 SynthID result** The Results and Accessibility CSU shall display the SynthID finding as a separate labeled result.
- 5.3.1.25 Unsupported SynthID explanation** The Results and Accessibility CSU shall display “not applicable” when SynthID does not support the item type.
- 5.3.1.26 Signal states** The Results and Accessibility CSU shall use the signal-state labels defined in Section 5.1.1: detected, not detected, verified AI assertion, verified credential without an AI assertion, missing credential, invalid credential, not applicable, unavailable, and error, as applicable to each method.
- 5.3.1.27 Missing signal meaning** The Results and Accessibility CSU shall state that an unavailable or undetected signal does not establish human authorship.
- 5.3.1.28 Warning evidence** The Results and Accessibility CSU shall identify the evidence used to trigger the overall warning.

5.3.1.29 Inconclusive explanation The Results and Accessibility CSU shall show an inconclusive state when at least one check completes but the available evidence does not meet the warning rule.

5.3.1.30 Source context The Results and Accessibility CSU shall show the scan time and source page for the item.

5.3.1.31 Detection limits The Results and Accessibility CSU shall display "AI detection can be wrong. Missing credentials or watermarks do not prove that a person created this content."

5.3.1.32 Origin and truth The Results and Accessibility CSU shall not describe an item as a scam or misinformation solely because it may be AI generated.

5.3.1.33 All item explanations The Results and Accessibility CSU shall list every analyzed item with a control that opens its explanation, including items without an inline warning.

5.3.1.34 Copy result The Results and Accessibility CSU shall provide a Copy action for an individual item result.

5.3.1.35 Copied fields The Results and Accessibility CSU shall place the item type, source page, scan time, overall state, and separate available findings on the clipboard when Copy is activated.

5.3.1.36 Copy confirmation The Results and Accessibility CSU shall display a "Copied" confirmation after an individual item result is successfully copied to the clipboard.

5.3.1.37 Copy failure The Results and Accessibility CSU shall display a "Copy failed" error message when an individual item result cannot be copied because clipboard access fails.

5.3.1.38 Share action The Results and Accessibility CSU shall provide a Create share link action for an individual item result.

5.3.1.39 Share preview The Results and Accessibility CSU shall show the information that will appear in the shared report before the user creates a link.

5.3.1.40 Copy link The Results and Accessibility CSU shall provide a Copy link action after a URL is created.

5.3.1.41 Permission scope The Popup and Controls CSU shall restrict required permissions to activeTab, scripting, storage, clipboardWrite, and host access to the RealEye HTTPS service, with HTTP and HTTPS image origins declared only as optional host permissions.

5.3.1.42 Upload disclosure The Extension Interface CSC shall disclose that selected or scanned content is sent to the analysis service before the first scan.

5.3.1.43 No unsolicited upload The Extension Interface CSC shall not transmit page content before the user starts a scan.

5.3.1.44 Service failure The Extension Interface CSC shall display a "Service unavailable" error message when the analysis service cannot be reached.

5.3.1.45 Retry The Extension Interface CSC shall offer a retry action following a recoverable network failure.

5.3.1.46 Failure distinction The Extension Interface CSC shall display "Failed" and "Inconclusive" as separate result states.

5.3.1.47 Color labels The Extension Interface CSC shall use text labels with all warning colors.

5.3.1.48 Keyboard controls The Extension Interface CSC shall expose scan choices, result items, and share controls to keyboard focus.

5.3.1.49 Visible focus The Extension Interface CSC shall provide visible keyboard focus on its interactive controls.

5.3.1.50 Accessible names The Extension Interface CSC shall expose an accessible name describing the function of each icon-only control.

5.3.1.51 Status announcement The Extension Interface CSC shall announce scan completion and failure in an accessible status region.

5.3.1.52 Permission purpose The help view shall list each requested browser permission with the feature that uses it.

5.3.1.53 Unsupported input feedback The Results and Accessibility CSU shall display the applicable language, length, format, or size limit when an item is rejected.

5.3.1.54 Share lifetime disclosure The Results and Accessibility CSU shall disclose before link creation that the report is public to link holders for seven days and cannot be deleted early through the baseline interface.

5.3.1.55 Scan cap disclosure The Popup and Controls CSU shall display the maximum of 100 text blocks and 50 images before a full-page scan.

5.3.1.56 Image permission prompt The Popup and Controls CSU shall list the additional image origins requiring access before presenting an "Allow image access" action.

5.3.1.57 User granted image access The Popup and Controls CSU shall request optional host access only for the listed image origins when the user activates "Allow image access".

5.3.1.58 Permission refusal The Results and Accessibility CSU shall identify images skipped because origin permission was denied or revoked.

5.3.2 Page Interaction CSC

The Page Interaction CSC will inspect supported text and images within the user-selected scan scope. It will associate findings with the corresponding webpage content, display inline warnings, and allow users to open explanations or dismiss warnings while continuing to use the page.

5.3.2.1 User initiated inspection The Page Interaction CSC shall inspect the active page only after a user initiates a scan.

5.3.2.2 Visible text discovery The Selection and Extraction CSU shall enumerate extractable visible text blocks within the active page's currently loaded content for a full-page scan.

5.3.2.3 Image discovery The Selection and Extraction CSU shall enumerate supported images within the active page's currently loaded content for a full-page scan.

5.3.2.4 Exclude extension text The Page Interaction CSC shall exclude text belonging solely to extension-generated controls from content extraction.

5.3.2.5 Item identity The Page Interaction CSC shall associate each extracted item with its content type, page location, and scan identifier.

5.3.2.6 Empty text removal The Page Interaction CSC shall remove empty or whitespace-only text items before submitting text for analysis.

5.3.2.7 Text deduplication The Selection and Extraction CSU shall not submit identical whitespace-normalized text more than once within the same scan.

5.3.2.8 Image deduplication The Selection and Extraction CSU shall not submit the same resolved image source URL more than once within the same scan.

5.3.2.9 Unreadable image The Page Interaction CSC shall report an unsupported image separately when its bytes cannot be retrieved or decoded.

5.3.2.10 Inaccessible frame The Page Interaction CSC shall report an inaccessible frame separately when Chrome does not permit extraction from that frame.

5.3.2.11 Frame coverage The Page Interaction CSC shall not treat content in an inaccessible frame as scanned.

5.3.2.12 Scan context The Page Interaction CSC shall capture the page URL and time of scan for display with the result.

5.3.2.13 Text selection The Page Interaction CSC shall identify the selected text range when the user requests a selected content scan with highlighted text.

5.3.2.14 Text scope The Page Interaction CSC shall limit a selected text scan to the highlighted text rather than scanning the full page.

5.3.2.15 Image selection The Page Interaction CSC shall identify a user-chosen image when the user requests a selected image scan.

5.3.2.16 Image scope The Page Interaction CSC shall limit a selected image scan to the chosen image rather than scanning the full page.

5.3.2.17 Missing selection The Page Interaction CSC shall ask the user to select text or an image if “Selected content” is chosen without a valid selection.

5.3.2.18 Empty scope The Page Interaction CSC shall display “No supported content found” when no supported items are found in the chosen scan scope.

5.3.2.19 Result association The Page Interaction CSC shall keep each analysis result associated with the correct page item for annotation.

5.3.2.20 Removed content The Page Annotation CSU shall discard an item result when its associated page item can no longer be located.

5.3.2.21 Warning annotation The Page Interaction CSC shall mark a supported item when its result meets the configured warning rule.

5.3.2.22 Distinct indicators The Page Annotation CSU shall outline a flagged item with a border and a "Possible AI" text label.

5.3.2.23 Readable content The Page Annotation CSU shall place warning labels outside the content bounds of the flagged item.

5.3.2.24 Open explanation The Page Interaction CSC shall provide a control on an inline warning that opens the corresponding item explanation.

5.3.2.25 Dismiss indicator The Page Interaction CSC shall provide a “Dismiss” control on each inline warning that removes that warning for the current scan.

5.3.2.26 Replace indicators The Page Interaction CSC shall remove previous scan indicators before displaying indicators from a replacement scan.

5.3.2.27 Webpage access The Page Annotation CSU shall leave webpage scrolling, text selection, and link activation operable while scan results are displayed.

5.3.2.28 Limit notice The Selection and Extraction CSU shall display a scan-limit notice when additional eligible items exceed the 100-text-block or 50-image cap.

5.3.2.29 Text block formation The Selection and Extraction CSU shall form text items from visible p, li, h1-h6, blockquote, pre, and figcaption elements without repeating descendant text already assigned to an item.

5.3.2.30 Text normalization The Selection and Extraction CSU shall collapse consecutive whitespace into one space and trim leading and trailing whitespace before applying text limits.

5.3.2.31 Scan order The Selection and Extraction CSU shall select eligible items in document order within each content type.

5.3.2.32 Scan caps The Selection and Extraction CSU shall limit each scan to the first 100 eligible text blocks and first 50 eligible images.

5.3.2.33 Short or long text The Selection and Extraction CSU shall mark text outside the 200-to-5000-character range as skipped without truncating it.

5.3.2.34 Batch limits The Selection and Extraction CSU shall divide eligible items into requests that satisfy the per-request limits in Section 5.5.3.

5.3.2.35 Original image retrieval The Selection and Extraction CSU shall retrieve original image bytes through an extension service worker when host access to the image origin is granted.

5.3.2.36 Permission boundary The Selection and Extraction CSU shall omit image retrieval when the required origin permission is absent.

5.3.2.37 Original asset preservation The Selection and Extraction CSU shall submit original image bytes without screenshot conversion or recompression.

5.3.2.38 Redirect permission The Selection and Extraction CSU shall stop an image fetch that redirects to an origin lacking granted host access.

5.3.2.39 Fetch scope The Selection and Extraction CSU shall accept image-fetch requests only for image URLs associated with the user-initiated scan in the active tab.

5.3.3 Server CSC

The Server CSC will analyze submitted text and images using classification and available provenance and watermark evidence. It will return separate findings, overall result states, and explanations, report unavailable checks or analysis failures, and provide access to user-approved shared reports.

5.3.3.1 Scan request The Analysis API CSU shall accept a request containing a scan identifier and one or more supported items.

5.3.3.2 Type validation The Analysis API CSU shall validate each submitted item type before invoking detection.

5.3.3.3 Empty request rejection The Analysis API CSU shall reject a request that contains no supported items with an identifiable error response.

5.3.3.4 Item identifiers The Analysis API CSU shall assign or preserve a unique identifier for each analyzed item.

- 5.3.3.5 Text routing** The Analysis API CSU shall route a text item to the text classifier.
- 5.3.3.6 Image routing** The Analysis API CSU shall route an image item to the image classifier.
- 5.3.3.7 C2PA coordination** The Analysis API CSU shall request C2PA verification for a JPEG, PNG, or WebP item when the original, unmodified file bytes are supplied within the image limits in Section 5.5.3.
- 5.3.3.8 SynthID coordination** The Analysis API CSU shall request a SynthID check only when an accessible detector supports the submitted item.
- 5.3.3.9 Classifier independence** The Analysis API CSU shall run the applicable classifier even when no provenance or watermark information is available.
- 5.3.3.10 Partial evidence** The Analysis API CSU shall preserve the result of a completed check when another check is unavailable or fails.
- 5.3.3.11 Independent findings** The Analysis API CSU shall return independent status and evidence fields for each applicable detection method.
- 5.3.3.12 Inapplicable method** The Analysis API CSU shall return an explicit “not applicable” status for a method that does not support an item type.
- 5.3.3.13 Unavailable method** The Analysis API CSU shall return an explicit “unavailable” status when a method cannot be accessed for a supported item.
- 5.3.3.14 Check error** The Analysis API CSU shall return an explicit error status when a requested check fails during processing.
- 5.3.3.15 Response association** The Analysis API CSU shall return item results associated with their request item identifiers.
- 5.3.3.16 Rule version** The Analysis API CSU shall include the applied warning rule version in the scan response.
- 5.3.3.17 Missing evidence** The Analysis API CSU shall preserve missing or unavailable signals as non-voting statuses in the response.
- 5.3.3.18 Partial scan** The Analysis API CSU shall allow an extension scan to end with partial results when at least one item was analyzed.
- 5.3.3.19 Total failure** The Analysis API CSU shall return an error response indicating that no submitted item could be analyzed when all submitted items fail analysis.
- 5.3.3.20 Text input** The text inference module shall accept supported text extracted from a scan item.
- 5.3.3.21 Text estimate** The text inference module shall return an AI likelihood estimate within the inclusive range 0 to 1 for a completed text analysis.
- 5.3.3.22 Text label** The text inference module shall return a model class label associated with the estimate.
- 5.3.3.23 Unsupported text handling** The text inference module shall return no numeric estimate for text rejected under the language or length limits in Section 5.5.3.
- 5.3.3.24 Image input** The image inference module shall accept a supported decoded image extracted from a scan item.
- 5.3.3.25 Image estimate** The image inference module shall return an AI likelihood estimate within the inclusive range 0 to 1 for a completed image analysis.

- 5.3.3.26 Image label** The image inference module shall return a model class label associated with the estimate.
- 5.3.3.27 Corrupt image** The image inference module shall report an unsupported or corrupt image without producing a fabricated estimate.
- 5.3.3.28 Model version** The Text and Image Inference CSU shall identify the version of the model artifact used for inference.
- 5.3.3.29 Consistent preprocessing** The Text and Image Inference CSU shall apply the same inference preprocessing used in its recorded evaluation.
- 5.3.3.30 Estimate field** The Analysis API CSU shall return a numeric model estimate only when classification succeeds.
- 5.3.3.31 Estimate metadata** The Analysis API CSU shall return `estimate_kind="model_estimate"` for every uncalibrated classifier score.
- 5.3.3.32 Missing estimate** The Analysis API CSU shall not replace a missing classifier estimate with zero.
- 5.3.3.33 Mixed content results** The Text and Image Inference CSU shall return separate text and image model results for a mixed-content scan.
- 5.3.3.34 Original asset inspection** The credential verification module shall attempt to inspect provenance data on a supported original image asset when its data is accessible.
- 5.3.3.35 Credential integrity** The credential verification module shall verify the integrity of an available credential before presenting it as verified.
- 5.3.3.36 AI provenance assertion** The credential verification module shall report whether a verified credential records AI generation when that assertion is available.
- 5.3.3.37 Other provenance** The credential verification module shall distinguish a verified credential that lacks an AI generation assertion from one that records AI generation.
- 5.3.3.38 Missing and invalid credentials** The credential verification module shall distinguish missing credentials from failed credential verification.
- 5.3.3.39 Inaccessible asset** The credential verification module shall report “unavailable” when the original asset required for a check cannot be accessed.
- 5.3.3.40 Absent credential meaning** The Warning Interpretation CSU shall exclude absent C2PA credentials from evidence of human origin.
- 5.3.3.41 Detector access** The watermark availability and detection module shall test whether an authorized detector is accessible for the item type before attempting a check.
- 5.3.3.42 Unsupported watermark type** The watermark availability and detection module shall report “not applicable” for an unsupported item type.
- 5.3.3.43 Inaccessible SynthID detector** The watermark availability and detection module shall report “unavailable” when a detector for a supported item type cannot be accessed.
- 5.3.3.44 Positive watermark** The watermark availability and detection module shall report a positive watermark finding only when the accessible detector returns a positive result.
- 5.3.3.45 Negative watermark** The watermark availability and detection module shall report a negative finding only when a supported check actually completes without detection.

- 5.3.3.46 Watermark error** The watermark availability and detection module shall preserve a detector error separately from a negative finding.
- 5.3.3.47 Evidence source** The C2PA and Accessible SynthID CSU shall return the method name and status for each provenance or watermark finding.
- 5.3.3.48 Signal coverage** The C2PA and Accessible SynthID CSU shall return the limitation "Not all AI-generated content contains detectable credentials or watermarks." with its findings.
- 5.3.3.49 Versioned interpretation** The Warning Interpretation CSU shall evaluate each completed item using a versioned warning rule.
- 5.3.3.50 Classifier threshold** The Warning Interpretation CSU shall produce Warning when an available classifier estimate is greater than or equal to its validated, modality-specific threshold.
- 5.3.3.51 Verified AI provenance** The Warning Interpretation CSU shall produce Warning when verified C2PA credentials assert AI generation, regardless of other signal results.
- 5.3.3.52 Positive watermark evidence** The Warning Interpretation CSU shall produce Warning when an authorized SynthID check returns a positive detection, regardless of other signal results.
- 5.3.3.53 Separate findings** The Warning Interpretation CSU shall retain separate findings even when multiple methods disagree.
- 5.3.3.54 Conflicting evidence** The Warning Interpretation CSU shall return a conflict flag when at least one warning trigger is present and a completed classifier estimate is below its threshold.
- 5.3.3.55 Overall result** The Warning Interpretation CSU shall return exactly one overall state per item using the precedence rules in Section 5.1.1.
- 5.3.3.56 Warning explanation** The Warning Interpretation CSU shall attach an explanation of the rule that produced an item warning state.
- 5.3.3.57 Separate overall state** The Warning Interpretation CSU shall return the model estimate and overall state in separate response fields.
- 5.3.3.58 Threshold version** The Warning Interpretation CSU shall record the configured threshold values in the warning rule version used for the scan.
- 5.3.3.59 Public report link** The Sharing and Report Viewer CSU shall create a report snapshot readable by anyone possessing its valid link without a recipient account.
- 5.3.3.60 Report URL** The Sharing and Report Viewer CSU shall return a URL associated with that report snapshot.
- 5.3.3.61 Report fields** The report access module shall display the item type, source page, scan time, overall state, and separate available findings.
- 5.3.3.62 Shared limitations** The report access module shall display the detector limitations alongside the shared findings.
- 5.3.3.63 Expired report** The report access module shall show an unavailable or expired state for a link whose report no longer exists.
- 5.3.3.64 Manual distribution** The Sharing and Report Viewer CSU shall not send a message to another person on the user's behalf.

5.3.3.65 Individual report scope The Sharing and Report Viewer CSU shall not include unrelated items from the page in an individual item report.

5.3.3.66 Private logs The Analysis API CSU shall omit submitted text, image bytes, source URLs, and share tokens from application and access logs.

5.3.3.67 Explicit report storage The Sharing and Report Viewer CSU shall create a stored report only after the user chooses Create share link.

5.3.3.68 Request size The Analysis API CSU shall reject a request body exceeding 16 MiB with HTTP 413 before inference.

5.3.3.69 Request item limit The Analysis API CSU shall reject a request containing more than 10 items with HTTP 422.

5.3.3.70 Text length validation The Analysis API CSU shall return an item-level unsupported status for text outside 200 to 5000 normalized Unicode code points.

5.3.3.71 Text language validation The Analysis API CSU shall return an item-level unsupported status when the released language detector does not identify the text as English.

5.3.3.72 Image limits The Analysis API CSU shall reject an image item outside the format, byte, dimension, or pixel limits in Section 5.5.3 before model inference.

5.3.3.73 Rate limiting The Analysis API CSU shall reject requests exceeding 60 analysis requests per source IP in a rolling 60-second window with HTTP 429 and a Retry-After value.

5.3.3.74 Browser origin policy The Analysis API CSU shall reject browser requests carrying an Origin outside the configured release extension origin with HTTP 403.

5.3.3.75 Privileged access The Analysis API CSU shall reject administrative or configuration requests without a valid server-side administrator credential with HTTP 401.

5.3.3.76 Secret isolation The Analysis API CSU shall keep provider and administrative credentials exclusively in server-side configuration.

5.3.3.77 No early deletion The Sharing and Report Viewer CSU shall provide no user-facing early-deletion operation for a shared report in this baseline.

5.3.3.78 Report content minimization The Sharing and Report Viewer CSU shall store only item type, source page URL, scan time, overall state, rule and model versions, available findings, and detector limitations in a report snapshot.

5.3.3.79 Unpredictable links The Sharing and Report Viewer CSU shall use a cryptographically random report identifier with at least 128 bits of entropy.

5.3.3.80 Transient content cleanup The Analysis API CSU shall release submitted content and non-shared analysis results within 60 seconds after completion, cancellation, or timeout.

5.3.3.81 No content disk storage The Analysis API CSU shall process submitted text and image bytes without writing them to persistent storage.

5.3.3.82 No remote asset fetch The Analysis API CSU shall analyze image bytes supplied in a request without fetching caller-provided image URLs.

5.3.4 Evaluation CSC

The Evaluation CSC will maintain labeled examples and separate training, validation, and test datasets. It will prepare and evaluate text and image classifiers, document performance and limitations, and record the model versions and thresholds used to determine readiness for release.

5.3.4.1 Text labels The Evaluation CSC shall maintain separate labeled examples for human- authored and AI-generated text.

5.3.4.2 Image labels The Evaluation CSC shall maintain separate labeled examples for human- created and AI-generated images.

5.3.4.3 Dataset permissions The Evaluation CSC shall identify the source and permitted use of each evaluation dataset.

5.3.4.4 Dataset partitions The Evaluation CSC shall separate training, validation, and held-out test examples before final evaluation.

5.3.4.5 Duplicate isolation The Evaluation CSC shall prevent exact duplicate items from appearing in more than one of the training, validation, and held-out test splits.

5.3.4.6 Label mapping The Evaluation CSC shall record the label mapping used by each classifier.

5.3.4.7 Base model record The Evaluation CSC shall record the base model and version used for each fine-tuned model.

5.3.4.8 Released artifacts The Evaluation CSC shall save an inference-ready artifact for each model included in a release.

5.3.4.9 Evaluation metrics The Evaluation CSC shall report precision, recall, F1, and a confusion matrix for each classifier on held-out test data.

5.3.4.10 Class counts The Evaluation CSC shall report the number of held-out examples in each class for each classifier.

5.3.4.11 Evaluation record The Evaluation CSC shall record the evaluation date, dataset version, and model version alongside each result.

5.3.4.12 Threshold selection The Evaluation CSU shall select each classifier warning threshold using validation data only.

5.3.4.13 Frozen warning policy The Evaluation CSU shall record the selected thresholds, evidence-combination rule, validation precision, recall, and false-positive rates in the versioned evaluation report before held-out testing.

5.3.4.14 Reproducible release The Evaluation CSU shall record the dependency lockfile hash, model identifiers, language-detector version, operating systems, hardware, browser build, test corpus, trial count, and warm-start state with each qualification run.

5.4 Performance Requirements by CSC

These are acceptance targets. Timing trials use Section 5.5.3 and include client and server time where applicable. Required image-origin permissions are pre-granted in timed trials; separate functional tests exercise approval, refusal, revocation, and redirects. Time spent waiting for a human permission decision is excluded from latency and failure-feedback measurements. Failed requests and timeouts count as unsuccessful trials. Model-quality targets apply to held-out data and do not establish accuracy on all internet content.

5.4.1 Scan Button Acknowledged Within 250 ms

The Extension Interface CSC shall acknowledge a scan button activation within 250 milliseconds in at least 95% of trials.

5.4.2 Up to 150 Items Enumerated Within 3 Seconds

The Page Interaction CSC shall enumerate up to 100 supported text blocks and 50 supported images within 3 seconds in at least 95% of trials on the reference page corpus.

5.4.3 Selected Item Result Within 10 Seconds

The Extension Interface and Server CSCs shall display a successful classifier result for a valid selected item within 10 seconds in at least 90% of trials.

5.4.4 Full Page Progress Within 5 Seconds

The Extension Interface CSC shall display a progress or partial-results update within 5 seconds of starting a full-page scan in at least 95% of trials.

5.4.5 Thirty Item Page Completed Within 30 Seconds

The Page Interaction and Server CSCs shall return successful classifier results for all eligible items on a page with at most 20 text blocks and 10 images within 30 seconds in at least 90% of trials.

5.4.6 Timed Out Evidence Reported Within 5.25 Seconds

The C2PA and Accessible SynthID CSU shall return Unavailable by 5.25 seconds after check initiation in at least 95% of trials when a check exceeds its five-second deadline, without suppressing a completed classifier result.

5.4.7 Canceled Results Removed Within 1 Second

The Extension Interface CSC shall stop displaying pending results from a canceled scan within one second in at least 95% of trials.

5.4.8 Text Classifier F1 of at Least 0.75

The released text inference module shall achieve AI-class F1 of at least 0.75 on a held-out set of at least 200 examples with at least 100 examples per class at the released warning threshold.

5.4.9 Image Classifier F1 of at Least 0.75

The released image inference module shall achieve AI-class F1 of at least 0.75 on a held-out set of at least 200 images with at least 100 images per class at the released warning threshold.

5.4.10 False Positive Rates Reported by Modality

The Evaluation CSU shall report false-positive rate as $FP / (FP + TN)$ separately for human-authored text and human-created images at the released thresholds.

5.4.11 Core Workflow Completed by Keyboard

The Extension Interface CSC shall support starting a selected-content scan, inspecting an item, copying its findings, and dismissing results using only a keyboard.

5.4.12 Contrast Ratios Meet WCAG 2.2 AA

The Extension Interface CSC shall provide contrast ratios of at least 4.5:1 for normal text and 3:1 for large text and non-text control boundaries in extension-owned UI.

5.4.13 Interactive Targets at Least 24 CSS Pixels

The Extension Interface CSC shall provide interactive target areas of at least 24 by 24 CSS pixels.

5.4.14 Controls Remain Accessible at 200 Percent Zoom

The Extension Interface CSC shall preserve access to scan and explanation controls at 200% browser zoom without overlap or clipped labels.

5.4.15 Shared Report Resolves Within 5 Seconds

The Sharing and Report Viewer CSU shall make a newly created share link resolve to its report within five seconds in at least 95% of trials.

5.4.16 Share Link Valid for Seven Days

The Sharing and Report Viewer CSU shall keep a share link valid for seven days from creation.

5.4.17 Expired Snapshot Deleted Within 24 Hours

The Sharing and Report Viewer CSU shall delete a stored snapshot within 24 hours after its seven-day validity period ends.

5.4.18 Five Concurrent Single Item Scans

The Analysis API CSU shall complete five concurrent valid single-item requests without overload responses or failed classification in each of 20 load-test rounds.

5.4.19 Failure Feedback Within 35 Seconds

The Extension Interface CSC shall display a network or service failure within 35 seconds after scan initiation when no successful result is available.

5.4.20 Text False Positive Rate at Most 10 Percent

The released text inference module shall achieve a false-positive rate of at most 10% on the held-out human-authored text examples at the same threshold used for Section 5.4.8.

5.4.21 Image False Positive Rate at Most 10 Percent

The released image inference module shall achieve a false-positive rate of at most 10% on the held-out human-created image examples at the same threshold used for Section 5.4.9.

5.5 Project Environment Requirements

The following values specify the required qualification environment and deployment interfaces. A dependency lockfile and evaluation record identify exact builds and model artifacts. Hardware allocations and services described here are requirements, not claims of existing access or measured performance.

5.5.1 Development Environment Requirements

Category	Requirement
Processor and RAM	64-bit processor with at least 4 logical cores; 16 GiB RAM.
Storage and display	At least 30 GiB free SSD storage; 1280 by 720 display or larger.
Operating system	Windows 11 or macOS for client development; Ubuntu 24.04 LTS for server qualification. Record Windows and macOS versions in the qualification profile.
Browser	Desktop Chrome supporting Manifest V3; qualification uses the Stable build installed on the first qualification date, recorded and held fixed for all trials.
Frontend	React 18, TypeScript 5, Node.js 22, and a committed package lockfile.
Backend	Python 3.11 and FastAPI 0.115; exact patch releases pinned in the release lockfile.
Browser and backend interfaces	Chrome Extension APIs compatible with Manifest V3; a REST API implemented with FastAPI for HTTPS communication between the extension and backend.
Model tooling	PyTorch 2.6 and Hugging Face Transformers 4.49; exact patches and model revisions pinned in the release lockfile.
Evidence tooling	C2PA verification library with its version and trust configuration pinned in the release manifest; authorized SynthID integration when accessible.
Version control	Git repository containing extension, API, report viewer, model configuration, and qualification records.
Training resource	Local or institution-provided CUDA GPU with at least 16 GiB VRAM and 32 GiB host RAM; scheduled cloud allocation if local capacity is unavailable.

5.5.1.1 The development environment shall retain version-controlled source for the extension, API, model configuration, and report viewer.

5.5.1.2 The development environment shall provide the software and hardware profile in the development tables.

5.5.1.3 The development environment shall retain versioned labeled datasets with separate training, validation, and test partitions.

5.5.1.4 The development environment shall provide valid, missing, and invalid C2PA credential fixtures.

5.5.1.5 The development environment shall provide controlled SynthID unavailable, not-applicable, error, positive, and negative fixtures isolated from production findings.

5.5.1.6 The release manifest shall identify exact dependency versions and model revisions for reproducible installation.

Machine-learning models and datasets: RealEye will use Python, PyTorch, and Hugging Face Transformers to load and fine-tune pretrained machine-learning models. Separate models and labeled datasets will be used for text and image classification. The text dataset will contain human-authored and AI-generated text, while the image dataset will contain human-created or captured images and AI-generated images. Each dataset will be divided into separate training, validation, and test partitions. The specific model and dataset selections will be documented before training, including their sources, versions, and permitted uses.

Resource justification and acquisition: fine-tuning may require GPU capacity beyond the client laptop. The team will request institutional GPU access or a budget-approved temporary cloud allocation for training. A continuously hosted inference GPU is not required. CPU inference will be profiled first against the unchanged latency and concurrency targets; any acceleration requires a confirmed funded allocation and a recorded qualification profile.

Failure to meet a target remains an acceptance failure until the deployment is improved or the SRS is formally revised. Dataset access will be obtained under the permitted terms before ingestion. The C2PA verification package will be installed with documented trust settings and test assets before evidence qualification.

SynthID access: the team will check the official Google SynthID documentation for a permitted detector covering the required medium, request access where required, and verify that programmatic integration is authorized. Access to a detector interface does not by itself establish API access. If no authorized integration is available, the live service returns Unavailable for supported but inaccessible checks; controlled fixtures validate behavior without being represented as live detection. Unsupported item types return Not applicable. Reference: <https://deepmind.google/models/synthid/>

5.5.2 Deployment and Execution Environment

Category	Requirement
Client	Desktop Chrome on Windows 11, macOS, or ChromeOS; 4 logical CPU cores; 8 GiB RAM; 1280 by 720 display; extension installation permission. Qualify performance on Windows 11 and macOS; other platforms require qualification before equivalent performance is claimed.
Server compute	Ubuntu 24.04 LTS; at least 4 vCPUs, 16 GiB RAM, and 30 GiB free disk. CPU inference is the initial deployment profile; optional accelerator only after profiling and funded allocation.
Hosting	HTTPS-accessible Linux host running the pinned FastAPI application with Uvicorn. Provider choice does not change the execution contract.
Report storage	SQLite 3 on a persistent server volume; one application instance for this baseline; scheduled expiry cleanup at least hourly.
Network	An active internet connection that allows HTTPS access to the RealEye analysis service and shared-report endpoints.
Release configuration	Versioned text and image artifacts, validated thresholds, warning-rule version, C2PA trust settings, detector availability, and release extension origin.

5.5.2.1 The extension shall execute in the recorded desktop Chrome qualification build with Manifest V3 enabled.

5.5.2.2 The extension shall use HTTPS for all API and report communication.

5.5.2.3 The Text and Image Inference CSU shall deploy the model artifacts identified in the evaluation report.

5.5.2.4 The C2PA and Accessible SynthID CSU shall provide credential verification for accessible supported original image bytes.

5.5.2.5 The watermark availability and detection module shall return Unavailable when an authorized detector for a supported type is inaccessible.

5.5.2.6 The Sharing and Report Viewer CSU shall serve report links through a reachable HTTPS endpoint.

5.5.2.7 The Sharing and Report Viewer CSU shall persist snapshots until their specified expiry and deletion deadlines.

5.5.2.8 The Extension Interface CSC shall explain when Chrome restrictions prevent access to a page or frame.

5.5.2.9 The Analysis API CSU shall accept core scan requests without a user account.

Security boundary: core analysis is an account-free public service. A browser Origin allowlist restricts browser callers but is not authentication of arbitrary clients. Rate limits and input validation apply to all callers;

administrative and provider credentials stay on the server. The baseline does not claim that a key embedded in the extension can establish exclusive extension access.

5.5.3 Input Limits and Reference Conditions

Category	Requirement
Text language	English only. The release manifest pins the language-identification model and its configuration.
Text length	200 to 5000 Unicode code points after whitespace normalization, inclusive; no silent truncation.
Image formats	JPEG, PNG, and static WebP; animated images are unsupported.
Image bounds	At most 10 MiB encoded bytes; width and height each 32 to 4096 pixels inclusive; at most 16,000,000 decoded pixels.
Scan limits	At most 100 text blocks and 50 images per scan, chosen in document order within each type; selected-content scans contain one item.
API request bounds	At most 10 items and 16 MiB total encoded HTTP request body per request; image transport overhead counts toward the limit.
API quota	60 analysis requests per source IP in a rolling 60-second window; all tabs sharing an IP share this quota.
Reference client	Windows 11 with Intel Core i5-1135G7, or macOS with Intel/Apple silicon (CPU and OS version recorded); 8 GiB RAM; 1920 by 1080 display; 100% zoom except zoom tests.
Reference server	Recorded host meeting Section 5.5.2, including exact CPU, RAM, and any accelerator; released models resident in memory; no training workload. Qualification record identifies CPU-only or accelerated execution.
Reference network	At least 50 Mbps downstream and 20 Mbps upstream; round-trip latency to server at most 100 ms; no artificial packet loss.
Timing protocol	20 trials per timing target, including injected check timeouts for Section 5.4.6, after one uncounted warm-up scan; 95% means at least 19 of 20, 90% means at least 18 of 20.
Corpus and conditions	Versioned local test-page corpus covering minimum, typical, and maximum permitted item sizes, plus supported and restricted content. Record all inputs and browser/server builds.
Warning thresholds	One threshold per classifier, selected on validation data and frozen with the rule version before held-out tests; the same threshold applies to F1 and false-positive measurement.
Report policy	Seven-day access; no user early deletion; no excerpts or thumbnails; stored fields limited to the report schema.
Analysis privacy	No persistent raw content storage; transient content and non-shared results released within 60 seconds after terminal scan state.

The 200-character minimum intentionally excludes headlines, short captions, and short selected snippets. Such items receive a length-limit explanation; the baseline does not claim short-text classification. Demonstration text inputs must satisfy this length limit. Reference trials measure warm service behavior; cold deployment startup is outside the timing targets. Full-page completion within 30 seconds is qualified at 20 text blocks and 10 images, while the larger 100/50 cap defines supported scan capacity. Limits do not imply that every capped scan must finish

within 30 seconds. Failures retain partial results and produce the specified feedback. Model results describe the held-out corpus, not an assurance across every language, generator, or website.

5.5.4 Deferred Extensions

Video and audio analysis, OCR, deepfake-specific detection, phishing classification, fact checking, classification feedback, mobile clients, offline inference, and an account-based Trusted Circle with invitations are outside this acceptance baseline. Scanning inaccessible Chrome pages or frames and automatic infinite scrolling are also excluded.